

Note 12.2 Vectors

Component Form

Definition of Vector

The **vector** represented by the directed line segment $\overrightarrow{AB}$ has the **initial point** A and the **terminal point** B and its length $|\overrightarrow{AB}|$. Two vectors are **equal** if they have the same length and direction. The vectors are usually written by uppercase boldface letter such as $\mathbf{v}$. And the vector whose initial point is the **origin** will be the representative in **standard position**.

Definition of Component Form

If $\mathbf{v}$ is a **two-dimensional** vector in the plane equal to the vector with initial point at the origin and terminal point (v_1, v_2) , then **the component form** of v is

$$\mathbf{v} = \langle v_1, v_2 \rangle$$

$\mathbf{v}$ is a **three-dimensional** vector in the plane equal to the vector with initial point at the origin and terminal point (v_1, v_2, v_3) , then **the component form** of v is

$$\mathbf{v} = \langle v_1, v_2, v_3 \rangle$$

And the numbers $v_1, v_2, \dots$ are the **components** of $\mathbf{v}$. And given the initial point $P(x_1, y_1, z_1)$ and the terminal point $Q(x_2, y_2, z_2)$, the component form of $\overrightarrow{PQ}$ is

$$\mathbf{v} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Definition of Magnitude

The **magnitude** or **length** of the vector $\mathbf{v} = \overrightarrow{PQ}$ is the nonnegative number

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

And the vector with length 0 is the **zero vector** $\mathbf{0}$ and it has **no specific direction**.

Vector Algebra Operations

Definition of Addition and Scalar Multiplication

Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$, $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ be vectors and k is any scalar, then

- **addition**:

$$\mathbf{u} + \mathbf{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$$

- scalar multiplication:

$$k\mathbf{u} = \langle ku_1, ku_2, ku_3 \rangle$$

Geometric Interpretation

For addition

We just need to place the initial point of one vector at the terminal point of another. And the sum is called the **resultant vector**.

For multiplication

If $k > 0$, the resultant vector will have the same direction as $\mathbf{u}$ and if $k < 0$ the direction is opposite.

For the difference $\mathbf{u} - \mathbf{v}$, it is defined by

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$$

Properties of Vector Operations

Let $\mathbf{x}, \mathbf{y}, \mathbf{z}$ be vectors and α, β be scalars.

- $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$
- $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$
- $\mathbf{x} + \mathbf{0} = \mathbf{x}$.
- $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$.
- $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}$
- $(\alpha + \beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}$
- $(\alpha\beta)\mathbf{x} = \alpha(\beta\mathbf{x})$
- $1 \cdot \mathbf{x} = \mathbf{x}$
- $0\mathbf{u} = \mathbf{0}$

And when three or more space vectors lies in the same plane, we say they are **coplanar** vectors.

Unit Vectors

A vector $\mathbf{v}$ of **length 1** is called a **unit vector**. The **standard unit vectors** are

$$\mathbf{i} = \langle 1, 0, 0 \rangle, \mathbf{j} = \langle 0, 1, 0 \rangle, \mathbf{k} = \langle 0, 0, 1 \rangle$$

The any vector $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ can be written as the **linear combination** of the standard unit vectors as

$$\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$$

And we can call the scalars v_1, v_2, v_3 as the **i-component, j-component, k-component**.

And if $\mathbf{v} \neq \mathbf{0}$, then $\frac{\mathbf{v}}{|\mathbf{v}|}$ is a unit vector in the direction of $\mathbf{v}$, called the **direction** of the nonzero vector $\mathbf{v}$. So the equation

$$\mathbf{v} = |\mathbf{v}| \frac{\mathbf{v}}{|\mathbf{v}|}$$

expresses $\mathbf{v}$ as **its length times its direction**.

Midpoint of a Line Segment

The **midpoint** M of the line segment joining points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is the point

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$